

My STEM Book Title

An Informative Subtitle

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Preface

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Chapter 1

Introduction

1.1 Background

This is sample text.

Historical Note

Sidebars should be short. They are implemented using `tcbox` and are breakable.

1.2 Definitions

Definition 1.1. A **ring** R is a set with two binary operations ...

Worked Example: Ideal in $\mathbb{Z}$

Let $I = 2\mathbb{Z}$. Show I is an ideal: closure and absorption by ring elements.

Tip

Always define notation clearly.

Info

The ring $\mathbb{Z}/n\mathbb{Z}$ has exactly n elements.

Warning

A ring need not be a field (invertibility is not automatic).

1.3 Exercises

Exercise 1.2. Prove that the intersection of two ideals is an ideal.

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